

AKROMID®

A3 1 S3 black (1139)

PA66-I

AKROMID® A3 1 S3 black (1139) is an unreinforced, impact modified Polyamide 6.6. It is characterised by a medium dry impact strength whilst maintaining good flowability. Furthermore, the material is heat stabilised and therefore perfectly suitable for connecting and fixing systems which are used at elevated temperatures in the automotive and electro industry. Its good flowability qualifies the material for processing in multi-cavity moulds. For new projects, please consider our successor grade A3 1 S3 schwarz (11392).

Features

heat stabilised 130 impact modified

Properties

Modulus	Strength	Impact
2.700 MPa	63 MPa	180 kJ/m ²

Mechanical Properties

Tensile modulus	1 mm/min d.a.m.	2700 MPa
ISO 527-2	1 mm/min conditioned	1300 MPa
Tensile stress at yield	50 mm/min d.a.m.	63 MPa
ISO 527-2	50 mm/min conditioned	45 MPa
Tensile strain at yield	50 mm/min d.a.m.	5,6 %
ISO 527-2	50 mm/min conditioned	> 20 %
Tensile strain at break	50 mm/min d.a.m.	> 35 %
ISO 527-2	50 mm/min conditioned	> 100 %
Flexural modulus	2 mm/min d.a.m.	2500 MPa
ISO 178		
Flexural strength	2 mm/min d.a.m.	90 MPa
ISO 178		
Flexural strain at break	2 mm/min d.a.m.	7,5 %
ISO 178		

Disclaimer

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Charpy impact strength

ISO 179-1/1eU

23°C d.a.m.	no break
23°C conditioned	no break
-30°C d.a.m.	no break
-30°C conditioned	no break

Charpy notched impact strength

ISO 179-1/1eA

23°C d.a.m.	17 kJ/m ²
23°C conditioned	95 kJ/m ²
-30°C d.a.m.	10 kJ/m ²
-30°C conditioned	13 kJ/m ²

Izod notched impact strength

ISO 180/1A

23°C d.a.m.	15 kJ/m ²
-20°C d.a.m.	12 kJ/m ²
-40°C d.a.m.	12 kJ/m ²

Thermal Properties

Temperature of deflection under load HDT/A

ISO 75

1,8 MPa	70 °C
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Temperature of deflection under load HDT/B

ISO 75

0,45 MPa	213 °C
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Melting temperature

ISO 11357-3

DSC, 10K/min	262 °C
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Coefficient of linear thermal expansion

ISO 11359-1/2

23°C to 80°C parallel	0,87 10 ⁻⁴ /K
23°C to 80°C transverse	0,87 10 ⁻⁴ /K

Flammability

Burning rate

UL 94

1,6 mm Wall thickness	HB Class
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GWFI

IEC 60695-2-12

1,6 mm Wall thickness	650 °C
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GWIT

IEC 60695-2-13

1,6 mm Wall thickness	675 °C
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Burning rate (<100 mm/min)

FMVSS 302

> 1 mm Thickness	+
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General properties

Density	23°C	1,1 g/cm³
ISO 1183		
Humidity absorption	70°C, 62% r.H.	2,0 - 2,2 %
ISO 1110		
Molding shrinkage	flow	1,3 - 1,5 %
ISO 294-4	transverse	1,5 - 1,7 %

Electrical Properties

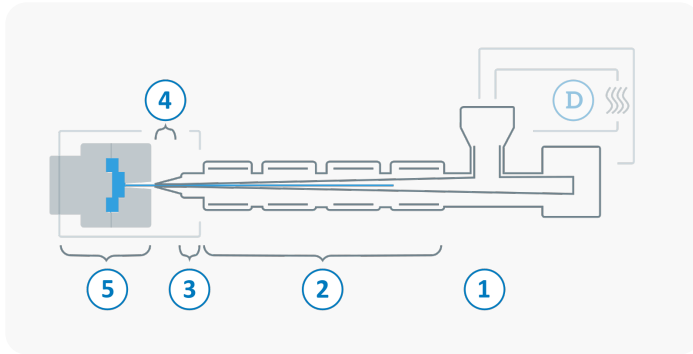
Volume resistivity IEC	d.a.m.	10¹⁵ Ω x cm
DIN EN IEC 62631-3-2		
Surface resistivity	d.a.m.	10¹⁴ Ω
DIN EN 62631-3-2		
Comparative tracking index	Test liquid A	600 V
IEC 60112		

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Processing

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only de-humidifying or vacuum dryers. Extensive drying can cause filling problem and surface defects.

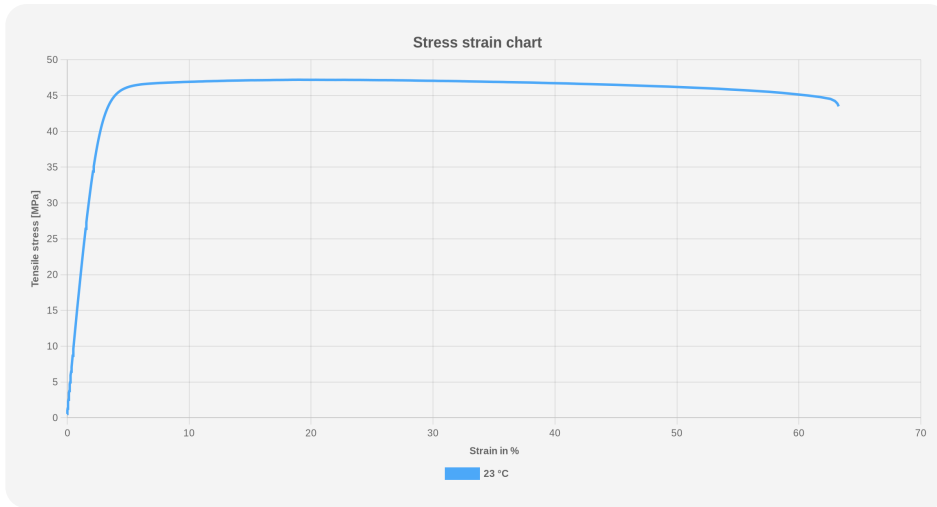


D	Drying time	2 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80 °C
	Processing moisture	0,02 - 0,1 %
1	Feed section	60 - 80 °C
2	Temperature Zone 1 - Zone 4	260 - 300 °C
3	Nozzle temperature	270 - 310 °C
4	Melt temperature	270 - 300 °C
5	Mold temperature	40 - 90 °C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

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Diagrams



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